

Mega Class: DP

Special class

✓✓ Live classes

✓✓ Doubts classes

✓✓ Assignment (Debug / quiz)

Mega cl

① LIS $\rightarrow$ DP $\rightarrow O(n^2)$

$\rightarrow O(n \log n)$ $\star$

②

Man. heights of

stacking

cuboids

$\star$

③

Edit Distance

Distance

④

Last stone weight $\underline{\underline{= 11}}$

① LIS

① $R \Sigma \Rightarrow \text{incl} / \Sigma_{\text{incl}}$

② $M_2 \rightarrow \underline{\text{DP} + \text{BS}}$

└ TD
└ BU
└ SO

✓
✓

10	9	2	5	3	7	101	18
0	1	2	3	4	5	6	7
✓	✓	✓	✓	✓	✓	✓	✓

$$0 \rightarrow \{10, 101\}$$

$$1 \rightarrow \{9, 101\}$$

$$2 \rightarrow \{2, 5, 7, 101\}$$

$$3 \rightarrow \{3, 7, 101\}$$

$$4 \rightarrow \{18,$$

]

0	1	2	3	4	5	6	7
10	9	2	5	3	7	10	18
10	9	2	5	3	7	10	18

~~5~~ ~~2~~
 Main = ~~2~~
 (4)

am

~~9~~
~~10~~ 2, ~~5~~₃, 7, ~~10~~₁₈

2 3 7 ~~10~~₁₈ 20

(5)

ans

10/9/2/5/3/10/10/8/18/20/23
0 1 2 3 4 5 6 7 8 9 10
↑ ↑ ↑ ↑ ↑ ↑ ↑ ↑ ↑ ↑

~~10~~, ~~9~~, ~~18~~, ~~10~~, ~~8~~, 23
2 3 18 20

ans-length

MaxL = ~~1~~ ~~2~~ ~~3~~ ~~4~~ 5

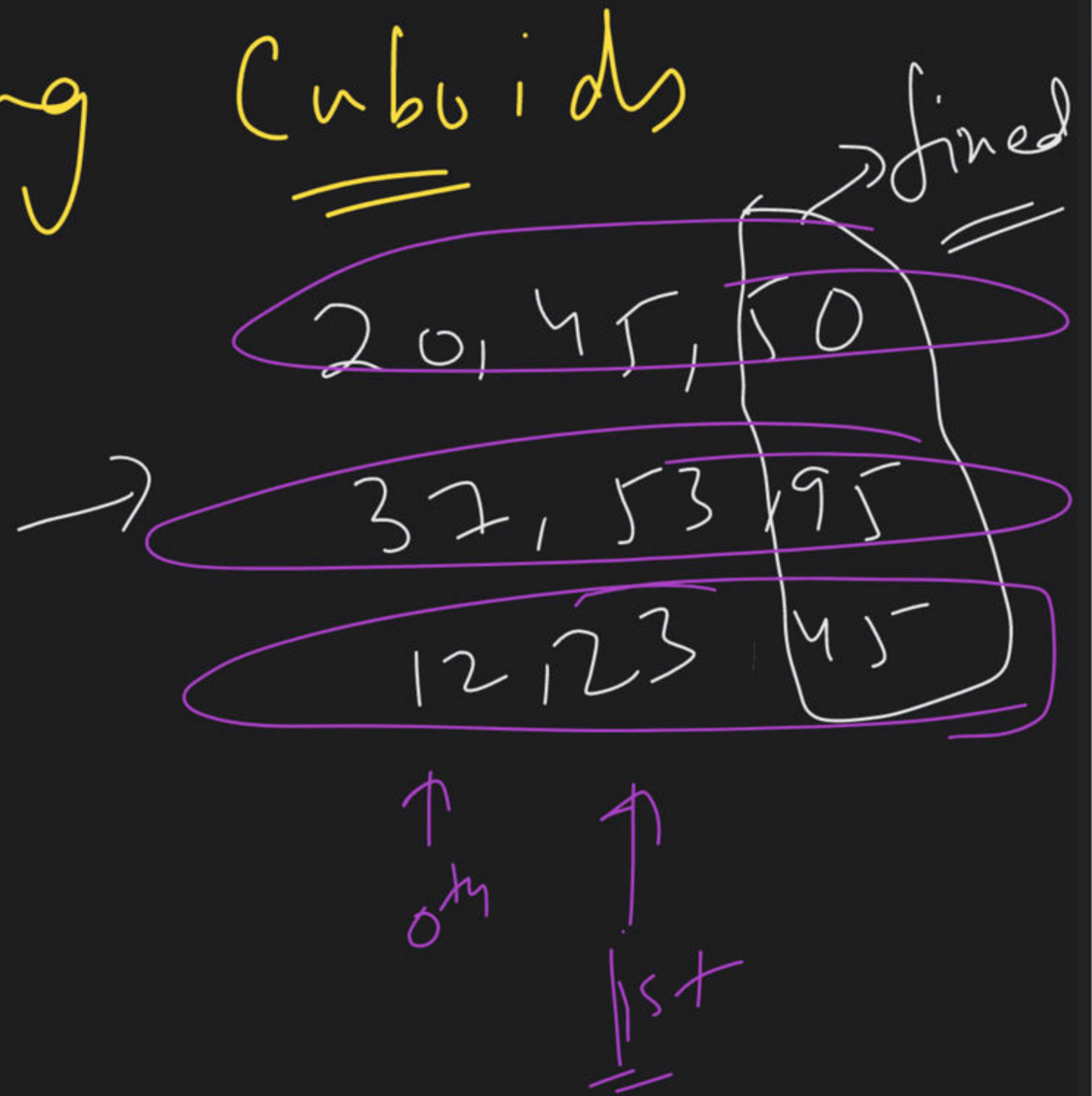
10 | 9 | 2 | 5 | 3 | 10 | 108 | 18
_{i i}

~~10~~, ³~~9~~, ~~10~~, 108
_{9 2 18}

final ans. length =

② Maximum ^{height} Stacking Cuboids

50, 45, 20
95, 37, 53
45 23 12



0 Based auto sorting

$v1 = [a, b, c]$ ↑ height

$v2 = [d, e, f]$ ↑ ~~height~~



→ 0th index on Sorting

① $a \neq d \rightarrow$ which is put first

$a == d \rightarrow$

↳ $b \neq e$ ke

hisab se

$b == e$

↳

c, f ke hisab se

1 Based surth

$V_1 \Rightarrow [a, b, c]$

$V_2 \Rightarrow [d, e, f]$

Compare

$\hookrightarrow$ 1-Based

$b, e \Rightarrow b \neq e$

$\hookrightarrow$ b, e hisab se.

$b == e$

$\hookrightarrow$ c, f ke hisab se

$c \neq f$

$\hookrightarrow$ Random

$\frac{V_1 \swarrow \searrow}{V_2}$

$V_1 = [13, 50, 50]$

$V_2 \rightarrow [18, 50, 50]$

0-Based $V_1 \rightarrow V_2$ | 1-Based $V_2 \rightarrow V_1$

[15, 10, 12]

→

[10, 12, 15]

[9, 13, 16]

→

[9, 13, 16]

[14, 5, 17]

→

[5, 14, 17]

[9, 13, 16]

[5, 14, 17]

[9, 13, 16]

[10, 12, 15]

[10, 12, 15]

5, 14, 17
10, 12, 15

~~5, 14, 17~~

5, 14, 17

[10, 12, 15]

15, 12, 10
16, 13, 9
17, 14, 5

10 | 12, 15
9, 13, 16 →
5, 14, 17
↓

③ 1049. Lat Stone weight 2

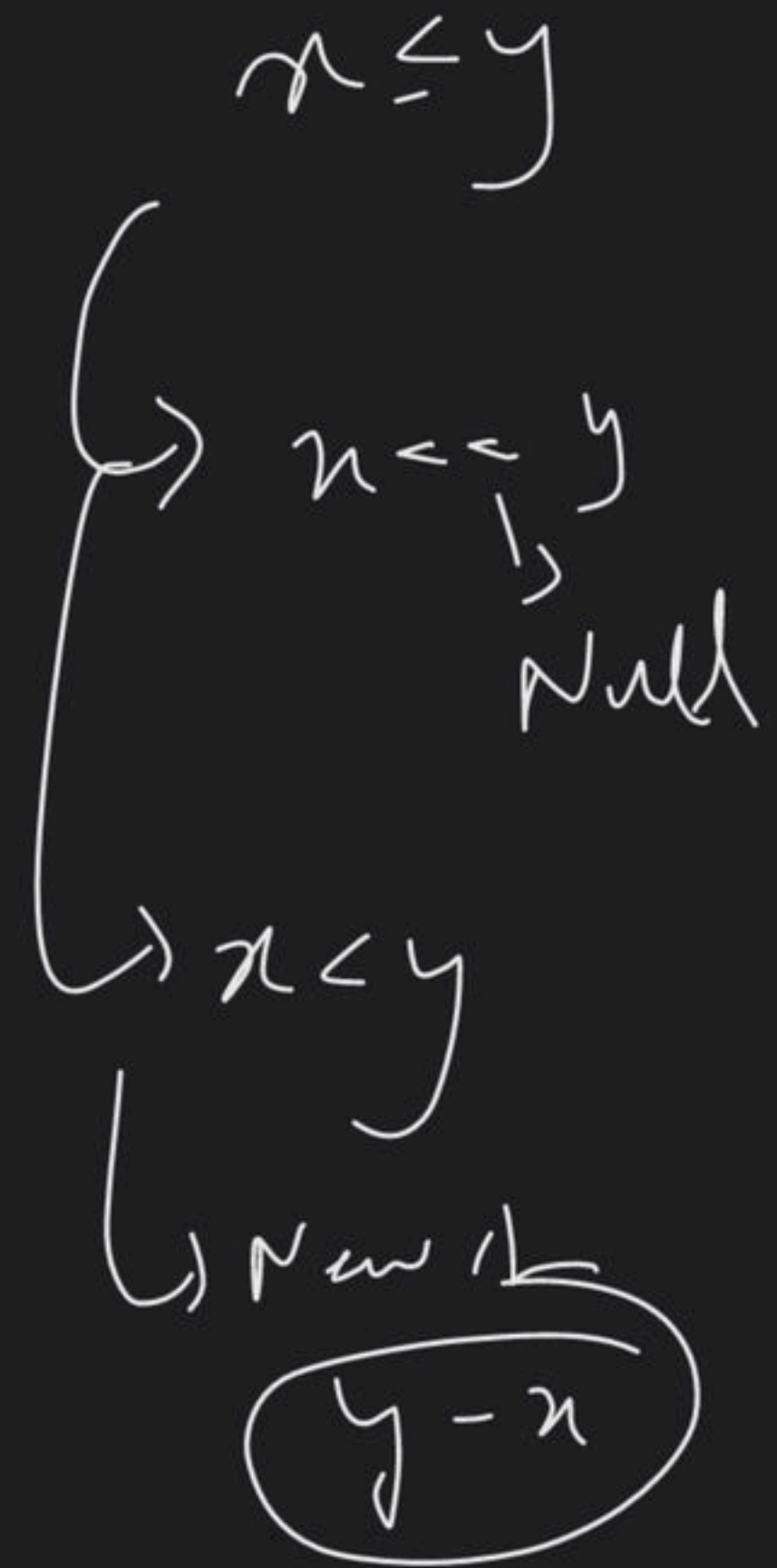
5 min → 10:25 PM

⁰ 2 | ¹ 7 | ² 4 | ³ 1 | ⁴ 8 | ⁵ 1

2 | 7 | 1 | 1 | 8 | 1

2 | 1 | 1 | 1 | 1

1 | 1 | 1 | 1 | 1
 ————
 1 → ans



2 | 7 | 4 | 1 | 1 | 8 | 1

✓

5 | 4 | 1 | 8 | 1

✓

✓

5 | 4 | 8

✓

1 | 8

→

7

1 | 1 | 2 | 4 | 7 | 8

✓
1

1 | 1 | 1 | 2 | 4

✓
1

1 | 1 | 1 | 2

✓
1

0

✓
1

13 | 1 | 2 | 4 | 7 | 8

1 | 2 | 4 | 7 | 8 | 13

✓
1

1 | 2 | 4 | 7

✓
1

1 | 2 | 2 | 4

✓
1

1 | 2 | 2

✓
1

①

21/26|31|33|40

— /

7|21|26|31

— /

5|7|21

5|14

— / (9

X

2 | 7 | 4 | 1 | 1 | 8 | 1

0 1 2 3 4 5

$$(-2 + 4)$$

$$(-7 + 8)$$

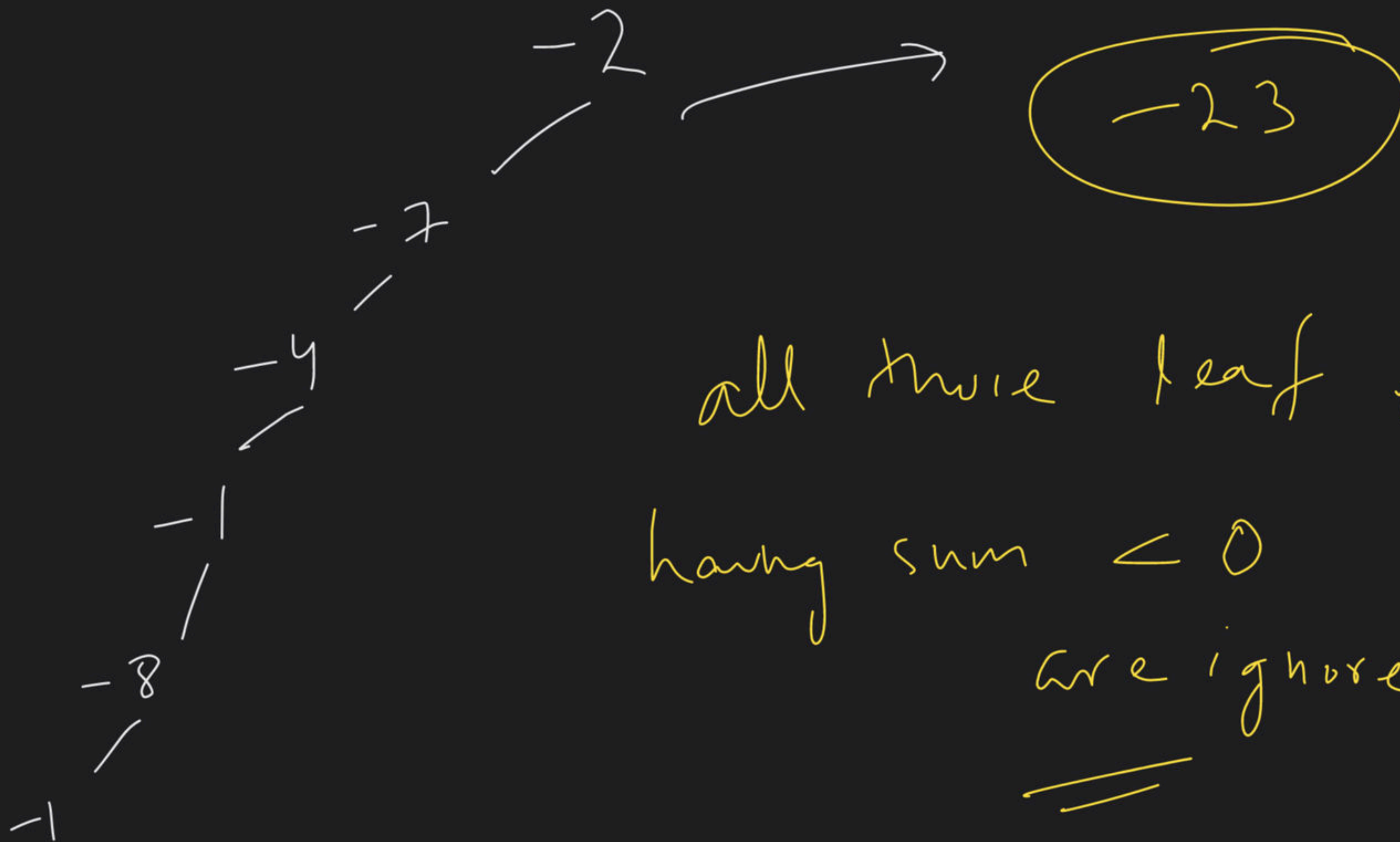
$$(-2 + 4 - 1)$$

$$(-7 + 8 - 1)$$

$$(-2 + 4 - 1 - 7 + 8 - 1)$$



$$(-2, -7, 4, -1, 8, -1)$$

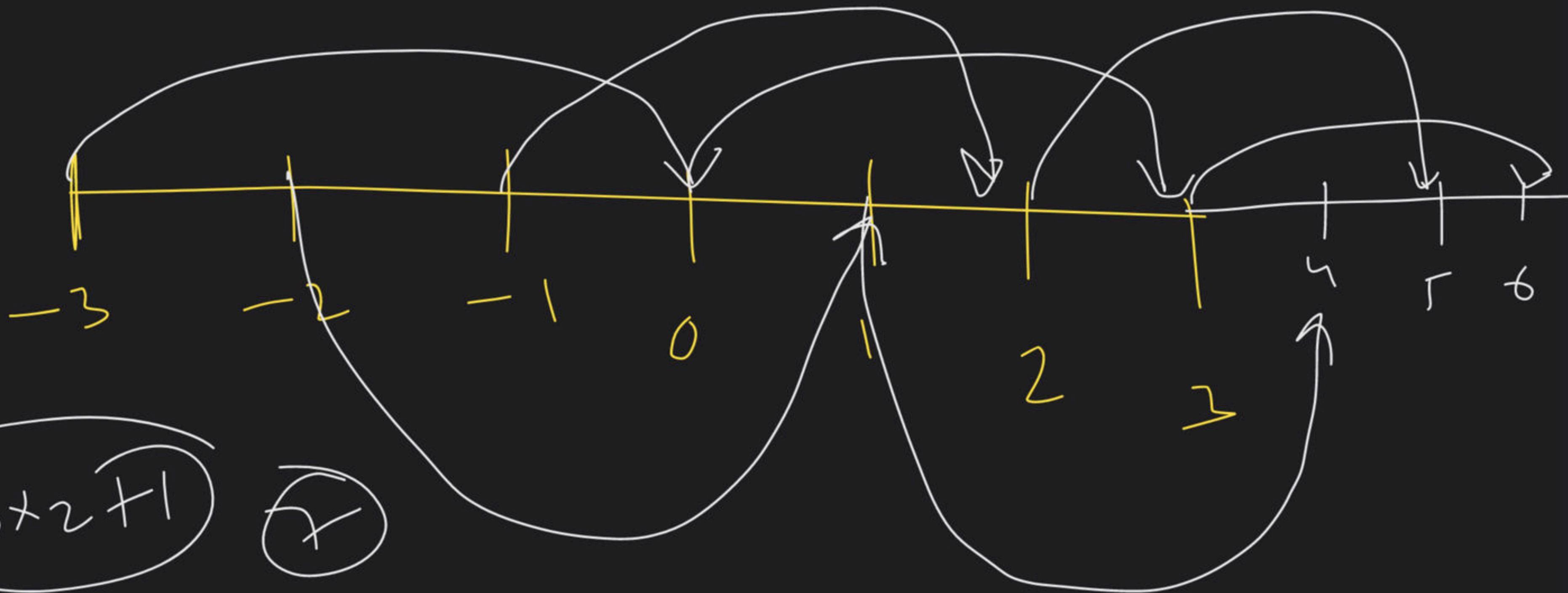


[1, 2]

0, 1, 2, 3, 4, 5, 6

7

-3 - +3



$3 \times 2 + 1$

7

[10, 10, 10, 10]
└┘

total → 40

40 + 10 → 50

Next 1.

~~Go~~

Current Q $\rightarrow$ diff way ?

2 | 7 | 4 | 1 | 8 | 1



$$|\sum G_1 - \sum G_2| \geq 0$$

minimized $|\sum G_1 - \sum G_2|$

-2 -7 4 -181

2 | 7 | 4 | 1 | 1 | 8 | 1 |

$G_1 \{2, 7, 1, 1\} \Rightarrow 11$

$G_2 \{4, 8\} \Rightarrow 12$

$|G_1 - G_2|$

$\downarrow$
1

ans

2 | 7 | 4 | 1 | 8 |

$G_1 \rightarrow (2, 7, 4, 1, 8)$

$G_2 \rightarrow (1) \rightarrow \dots \rightarrow 22 \rightarrow 1 \rightarrow \dots$
 $\hookrightarrow (2, 1)$